

Parallel Expectation-Maximization for Clustering

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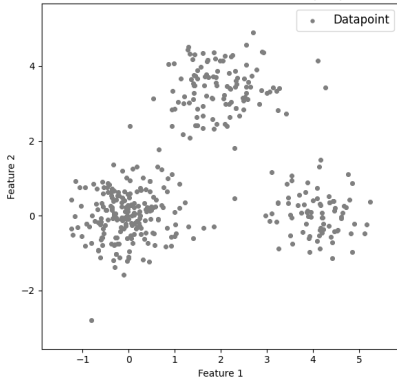


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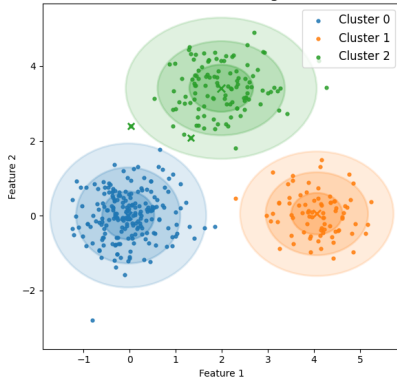
What is clustering - A Visual Example

Initial Dataset Visualization (2D)



First iteration of the clustering process

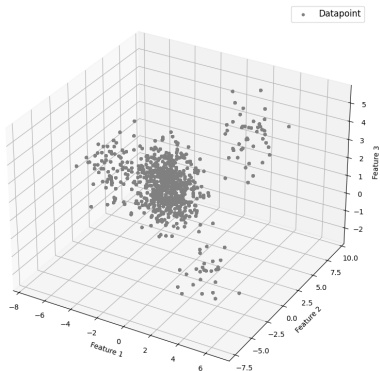
Final clustering



Final clustering result

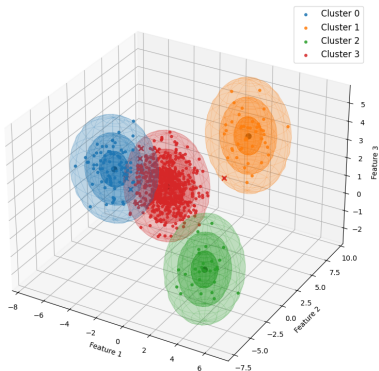
What is clustering - A Visual Example (3D)

Initial Dataset Visualization (3D)



First iteration (3D view)

Final Clustering



Final clustering (3D view)

What is clustering - More Formally

Gaussian Mixture Models

EM-Algorithm

E-Step

M-Step

Convergence

Computational Complexity Analysis

Parallel Design

Implementation

State-of-the-Art

Our Approach

Parallelization Strategy - E-Step

Parallelization Strategy - M-Step

Performance Considerations

Hybrid Parallelization

Scalability

Conclusion

Thank you!